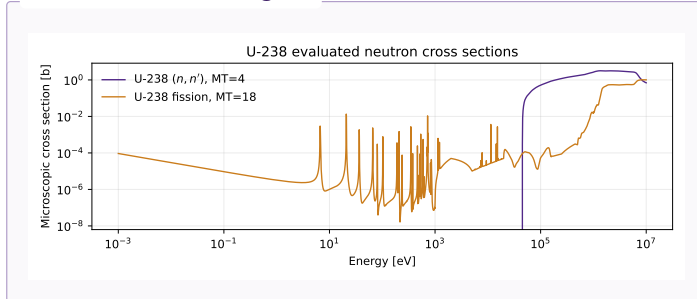


Prepare: Textbook §2.5

Primary objective. Estimate the number of elastic collisions needed to slow a neutron from E_0 to E_n for a target modeled by mass number $A \geq 1$.

1 Review: threshold reactions

Uranium-238 review figure



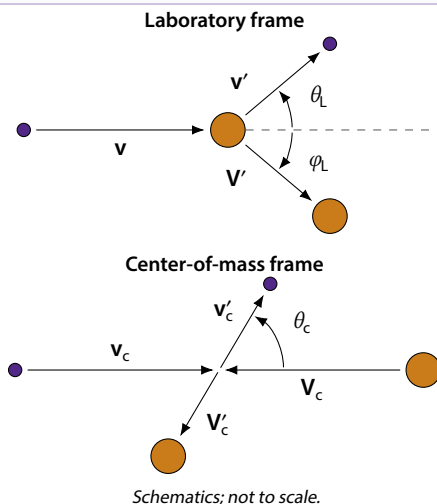
Review question. Do any of these interactions involve **threshold** reactions?

2 Frames and notation

The target is initially at rest in the **laboratory** (lab) frame. Primes denote quantities *after* the collision; a subscript c denotes the **center-of-mass** (CM) frame.

Keep the quantities distinct

m_n, M_N	Neutron and target masses; $A \equiv M_N/m_n \approx$ mass number.
$\mathbf{v}, \mathbf{V}$	Neutron and target velocities in the lab; $\mathbf{V} = \mathbf{0}$ initially.
v, V	Speeds: $v = \ \mathbf{v}\ , V = \ \mathbf{V}\ $.
$\mathbf{V}_{\text{CM}}$	Velocity of the CM frame relative to the lab.
E, E'	Incident and outgoing neutron energies in the lab.
μ_c, μ_L	$\mu_c = \cos \theta_c, \mu_L = \cos \theta_L$.
ϕ_L	Target recoil angle in the lab.



3 Conservation and frame changes

Conservation of momentum and energy in the **lab** frame requires

$$m_n \mathbf{v} = m_n \mathbf{v}' + M_N \mathbf{V}',$$

$$\frac{1}{2} m_n v^2 = \frac{1}{2} m_n v'^2 + \frac{1}{2} M_N V'^2.$$

In the **CM** frame, the corresponding statements are

$$m_n \mathbf{v}_c + M_N \mathbf{V}_c = \mathbf{0},$$

$$m_n \mathbf{v}'_c + M_N \mathbf{V}'_c = \mathbf{0},$$

$$\frac{1}{2} m_n v_c^2 + \frac{1}{2} M_N V_c^2 = \frac{1}{2} m_n v_c'^2 + \frac{1}{2} M_N V_c'^2.$$

Elastic scattering preserves the individual CM speeds:

$$v'_c = v_c, \quad V'_c = V_c.$$

The CM velocity and the transformations back to the lab are

$$\mathbf{V}_{\text{CM}} = \frac{m_n}{m_n + M_N} \mathbf{v} = \frac{1}{1 + A} \mathbf{v},$$

$$\mathbf{v}' = \mathbf{v}'_c + \mathbf{V}_{\text{CM}},$$

$$\mathbf{V}' = \mathbf{V}'_c + \mathbf{V}_{\text{CM}}.$$

Vector identities, such as $\mathbf{x} \cdot \mathbf{y} = xy \cos \theta$, connect energies and angles.

4 Elastic-scattering equations

Energy, angles, and the mean lab cosine

With E and A fixed, E' and θ_c share a one-to-one relationship:

$$\frac{E'}{E} = \frac{A^2 + 2A\mu_c + 1}{(A + 1)^2}.$$

The lab and CM angles are also linked one-to-one for $A \geq 1$ (apart from the endpoint noted below):

$$\mu_L = \frac{1 + A\mu_c}{\sqrt{A^2 + 2A\mu_c + 1}}.$$

For **isotropic CM scattering**, the average lab cosine is

$$\bar{\mu}_L \equiv \mathbb{E}[\cos \theta_L] = \frac{2}{3A} \quad (A \geq 1).$$

Endpoint clarification. At $A = 1$ and $\mu_c = -1$, $E' = 0$; a neutron at rest has no outgoing lab direction.

Ponderables

What values can E' be? μ_c ? μ_L ? What happens to $\bar{\mu}_L$ as A grows?

5 A quick summary of probabilities

Let X be a random variable with probability density $p_X(x)$ and support $x_L \leq x \leq x_U$.

A probability density

$$\int_{x_L}^{x_U} p_X(x) dx = 1,$$

$$p_X(x) \geq 0 \quad \text{for all } x \in [x_L, x_U],$$

and $p_X(x) = 0$ outside its support.

For $x_L \leq a \leq b \leq x_U$, the **probability** that $X \in [a, b]$ is

$$P(a \leq X \leq b) = \int_a^b p_X(x) dx.$$

The **expected value** (or “average”) of a function $f(X)$ is

$$\mathbb{E}[f(X)] \equiv \bar{f} = \int_{x_L}^{x_U} f(x) p_X(x) dx.$$

Here $\mathbb{E}[\cdot]$ denotes expectation; E denotes neutron energy.

6 Transforming probabilities

Probability densities can be transformed from one variable to another. We write $p_X(x)$ and $p_Y(y)$ for the densities of X and Y .

Suppose $Y = g(X)$ is a one-to-one, differentiable mapping. Corresponding probability elements satisfy

$$p_X(x) |dx| = p_Y(y) |dy|.$$

Change of variables

$$\begin{aligned} p_Y(y) &= p_X(g^{-1}(y)) \left| \frac{dg^{-1}(y)}{dy} \right| \\ &= \left| \frac{dx}{dy} \right| p_X(x) \Big|_{x=g^{-1}(y)} \\ &= \left| \frac{dy}{dx} \right|^{-1} p_X(x) \Big|_{x=g^{-1}(y)}. \end{aligned}$$

The support of Y is the image of the support of X under g .

Example: stretch and shift

Let X be uniform on $[0, 1]$ and let $Y = 2X - 1$. Then $p_X(x) = 1$ for $0 \leq x \leq 1$ (and zero elsewhere). Find the transformed density and support:

$$p_Y(y) = \left| \frac{dx}{dy} \right| p_X(x) = \underline{\hspace{2cm}}.$$

Support of Y : ; $p_Y(y) = 0$ otherwise.

Notes: transforming a density

7 Energy loss from collisions

Condition on an elastic collision that is isotropic in the CM frame. For incident neutron energy E , the outgoing-energy density is

Elastic-scattering energy distribution

$$p(E' | E) \equiv p(E \rightarrow E') = \begin{cases} \frac{1}{(1 - \alpha)E}, & \alpha E \leq E' \leq E, \\ 0, & \text{otherwise,} \end{cases}$$

FNRP 2.47

where

$$\alpha = \frac{(A - 1)^2}{(A + 1)^2}.$$

FNRP 2.48

Here $p(E' | E)$ is a *density in outgoing energy*, with E held fixed; it is not the probability that a collision occurs.

Example: expected outgoing energy

What is the expected outgoing energy $\bar{E}'$ of a neutron of energy E after an elastic collision with a target of mass ratio A ?

$$\bar{E}' = \mathbb{E}[E' | E] = \underline{\hspace{2cm}}.$$

8 Slowing-down decrement

The **average logarithmic energy loss**, also called the **slowing-down decrement**, is

Average logarithmic energy loss

$$\xi \equiv \mathbb{E} \left[\ln \frac{E}{E'} \mid E \right] = 1 + \frac{\alpha}{1 - \alpha} \ln \alpha.$$

FNRP 2.54 and 2.56

At $\alpha = 0$, the limiting value is $\xi = 1$.

The approximate number of elastic collisions needed to reduce the neutron energy from E_0 to E_n is

Collision-count estimate

$$n \approx \frac{1}{\xi} \ln \frac{E_0}{E_n}.$$

FNRP 2.59

Example: slowing down in water

How many collisions are required, on average, to slow a 2 MeV neutron to 1 eV in water?

Using the supplied effective, collision-weighted mixture value $\xi_{\text{water}} = 0.93$,

$$n \approx \frac{1}{0.93} \ln \left(\frac{2 \times 10^6 \text{ eV}}{1 \text{ eV}} \right) \approx \underline{\hspace{2cm}}.$$
